

EXP-009 — Does a contextual transformer retrieval encoder beat static vectors?

Production RAG v1 · evaluation-first · n = 20 questions / 22 evidence spans

STATUS — SUPPORTED ONCE THE TRUNCATION CONFOUND IS REMOVED

In the reference **256-token** configuration the preregistered accuracy bar was **not** met. The preregistered **512-token** sensitivity run clears it decisively and identifies the cause of movement in both directions: the encoder's **context window**, not its architecture.

0.775 best measured macro recall (D @512, was 0.475 baseline)	15/20 cases fully recalled (baseline 9/20)	+6 net cases vs BM25 with zero regressions	+4 net cases vs EXP-007C with zero regressions
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1. What changed in the environment

EXP-007's conclusion — "no transformer is reachable" — was **re-tested, not inherited**. Hugging Face and every embedding API are still blocked at the egress proxy (`403` at CONNECT, confirmed by the proxy's own failure log). But PyPI and an ONNX redistribution bucket *are* reachable, so a genuine transformer encoder was obtainable after all. EXP-009 was therefore **not blocked**, and no FastText fallback was used.

2. The encoder

`sentence-transformers/all-MiniLM-L6-v2` — a 6-layer BERT bi-encoder contrastively trained on ~1B sentence pairs explicitly for semantic search. 384 dimensions, fp32, unquantized (verified: zero quantization nodes in the ONNX graph). Attention-masked mean pooling, L2 normalized, cosine distance, **exact search with no ANN index**. No query or document prefix — the model is symmetric and defines no task instruction, so inventing one would be tuning.

Selected and committed in `experiments/EXP-009/model-preregistration.md` **before any retrieval result was observed**.

DECLARED PROVENANCE LIMITATION

Hugging Face is unreachable, so the bundle **cannot be checksummed against the upstream publisher**. Provenance rests on a third-party redistribution. Two independent checks are recorded instead: the architecture matches the published spec exactly, and a behavioural instrument check on held-out sentence pairs separates paraphrases from unrelated text by **0.468**. The decisive pair has *no content-word overlap at all* and still scores **0.603** — this encoder genuinely bridges vocabulary mismatch, which is what EXP-007's static vectors could not do.

3. Reproduction gates — nothing moved that should not have

cell	checks	verdict
A — BM25 (frozen baseline)	macro_span_recall 0.475 · cases_fully_recalled 9 · spans_found_at_10 10	PASS
B — FastText dense (EXP-007)	macro_span_recall 0.425 · cases_fully_recalled 8 · spans_found_at_10 9 · mrr 0.3599 · absent_at_300 5	PASS
E — BM25 + FastText RRF (EXP-007C)	macro_span_recall 0.6 · cases_fully_recalled 11 · spans_found_at_10 13	PASS

4. Primary result — reference 256-token window

cell	macro R	full	spans@10	doc R	MRR	a@10	a@50	a@300	ms
A — BM25 (frozen baseline)	0.475	9/20	10/22	0.825	0.280	12	3	1	560
B — FastText dense (EXP-007)	0.425	8/20	9/22	0.725	0.360	13	9	5	124
C — transformer dense	0.500	10/20	11/22	0.925	0.346	11	7	1	142
D — BM25 + transformer RRF	0.625	12/20	14/22	0.925	0.423	8	4	2	640
E — BM25 + FastText RRF (EXP-007C)	0.600	11/20	13/22	0.825	0.326	9	5	2	629

THE PRIMARY CONFIGURATION MISSED ITS OWN BAR

The preregistration required ≥ 0.10 (**two cases**) over FastText. C vs B came in at **+0.075** (1.5 cases) and D vs E at **+0.025** (one case). At $n = 20$ one case is 5 percentage points; no significance is claimed and a one-case difference is **not** reported as an improvement. This is reported as it fell.

5. Mechanism — the window, not the architecture

The reference window is 256 WordPiece tokens. Against the control chunks that means **35.2%** of chunks are truncated and the encoder sees only **51.3%** of all corpus tokens. The median truncated chunk is **745 tokens** — nearly three windows long. Splitting case movement by truncation explains *both* directions with one variable: the transformer wins wherever it can see the whole chunk, and loses wherever the answer sits past the cutoff.

6. The 512-token sensitivity run — mechanism confirmed

Declared in the preregistration **before any result was seen**, precisely so this could be tested rather than argued. Only the window changed: same weights, pooling, normalization, metric, chunks, queries and RRF parameters.

window	chunks truncated	corpus token coverage
256 (reference, primary)	35.2%	51.3%
512 (positional limit)	23.2%	76.1%

cell	macro R	full	spans@10	doc R	MRR	a@10	a@50	a@300	ms
A — BM25 (frozen baseline)	0.475	9/20	10/22	0.825	0.280	12	3	1	967
B — FastText dense (EXP-007)	0.425	8/20	9/22	0.725	0.360	13	9	5	670
C — transformer dense	0.575	11/20	13/22	0.925	0.339	9	4	1	160
D — BM25 + transformer RRF	0.775	15/20	17/22	0.925	0.449	5	3	2	512
E — BM25 + FastText RRF (EXP-007C)	0.600	11/20	13/22	0.825	0.326	9	5	2	443

All three reproduction gates still pass, and **cell E is unchanged at 0.600 / 11-of-20** — as it must be, since no transformer participates in it. That is the control showing the movement comes from the window and nothing else.

comparison	@256	@512	preregistered bar (≥ 0.10)
C vs B — transformer vs static	+0.075	+0.150	cleared
D vs E — fused, transformer vs static	+0.025	+0.175	cleared
D vs A — fused vs frozen baseline	+0.150	+0.300	—

ZERO-REGRESSION COMPARISONS — A FIRST FOR THIS PROJECT

At 512 the paired analysis reports **net +6 cases over BM25 with zero regressions**, and **net +4 over the EXP-007C fusion with zero regressions**. Rescued over BM25: AN-002, AN-004, AN-007, AN-008, AN-010, OA-006 . A comparison with no losing cases is qualitatively different from a net win.

OA-004, the fusion regression tracked since EXP-007, is rescued at 512.

WHAT THIS DOES NOT LICENSE

The 512 figure **does not retroactively become the primary result**. The preregistered primary was the reference 256 and it is reported as it fell. The 512 number rests on one run of $n = 20$, where one case is 5 percentage points, and 512 is the model's positional ceiling rather than a tuned value — there is no sweep here and none should be inferred.

7. Better on aggregate is not strictly better

The preregistration flagged this exact check. At 512 the transformer still loses **AN-002** and **AN-007** — two of the three cases FastText won. Regressed against FastText at 512: AN-002, AN-007, OA-008 .

case	group	A	B	C	D	E
AN-002	fasttext wins in exp007	27	1	77	49	2
AN-007	fasttext wins in exp007	18	2	22	7	4
AN-012	fasttext wins in exp007	47, 117	1, —	27, 189	4, —	2, —
AN-001	known failures	19	242	—	37	38

case	group	A	B	C	D	E
AN-003	known failures	—	—	91	—	—
AN-004	known failures	12	17	11	4	4
AN-006	known failures	29	63	217	56	56
AN-008	known failures	12	123	1	1	25
AN-010	known failures	49	—	64	95	97
AN-011	known failures	54	14	5	16	29
AN-012	known failures	47, 117	1, —	27, 189	4, —	2, —
OA-006	known failures	1, 29	18, —	1, 3	1, 8	2, 55
OA-004	fusion regression watch	5	73	56	14	17

Ranks of the expected evidence span at the 256-token configuration. "—" means absent from the top 300.

8. Reranker decision gate — counts only, no reranker was built

A reranker can only reorder what retrieval already returned; spans absent at pool depth are unreachable by any reranker. These are the counts that decision would rest on.

cell	1-10	11-30	31-50	51-100	101-300	absent@300	perfect-reranker ceiling @100
A — BM25 (frozen baseline)	10	7	2	1	1	1	0.909
B — FastText dense (EXP-007)	9	4	0	2	2	5	0.682
C — transformer dense	11	4	0	4	2	1	0.864
D — BM25 + transformer RRF	14	2	2	2	0	2	0.909
E — BM25 + FastText RRF (EXP-007C)	13	3	1	3	0	2	0.909

9. Defect found and fixed during this experiment

The first embedding build reported **0% truncation** on a corpus whose largest chunk is 6,857 tokens. `Tokenizer.from_file` restores the tokenizer's own saved 128-token truncation, so the "untruncated" reference length was itself truncated. **Stored vectors were never affected** — the encoding tokenizer's explicit override was always correct — but the measurement was wrong, and it would have put a false "no truncation" claim in this report. Fixed and recomputed.

10. Honest summary

- In the **reference 256-token configuration** the hypothesis **did not clear its own threshold** (+0.075 against a required +0.10).
- In the **preregistered 512-token sensitivity run** it clears the bar decisively: +0.150 dense, +0.175 fused, net +6 cases over BM25 with zero regressions.
- The transformer is **still not uniformly better**: AN-002 and AN-007 remain lost.
- The genuine finding is **mechanistic**: retrieval quality here is gated by how much of a chunk the encoder can see, and recall moves with that coverage in both directions.
- Reachability gains (absent@300 5 → 1, doc recall 0.725 → 0.925) are large enough not to be n = 20 noise, unlike the top-10 accuracy deltas.
- EXP-007's negative result is **superseded, not confirmed**: it was measured with an instrument this experiment shows to be the weaker one.

11. What is justified next

1. **Chunk the corpus to fit the encoder's window**, rather than chunking for its own sake. EXP-005 and EXP-008 both failed because they changed length without reference to any encoder's context limit; this is the first evidence that says what the limit should be.
2. **Still not a reranker**. The gate above shows what is and is not recoverable.
3. **Do not promote 512 on one run**. Re-measure before changing the frozen baseline.

Generated from experiments/EXP-009/*.json by scripts/build_exp009_pdf.py. Git commit adc73aea500c. Config hash caabf31b272a8d1c. Encoder sentence-transformers/all-MiniLM-L6-v2, exact cosine, no ANN index. Corpus snapshot snap_689e336380a054d8039dc35b2c09cd0a, chunk set cs_v1_control. Raw provider documentation is not redistributed.